Radiative transitions of strange heavy axial mesons in light-cone QCD sum rules

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Abstract

We study the radiative transitions $D_{s1}(2460) \to D_s\gamma$, $D_{s1}(2536) \to D_s\gamma$, and $B_{s1}(5830) \to B_s\gamma$ in light-cone QCD sum rules. The calculation keeps strange-quark-mass terms, hard photon emission, and two- and three-particle photon distribution-amplitude contributions. With the lattice-normalized photon distribution amplitude we obtain $\Gamma[D_{s1}(2460) \to D_s\gamma] = 16.1^{+4.4}_{-3.7}$ keV and $\Gamma[D_{s1}(2536) \to D_s\gamma] = 0.50^{+0.60}_{-0.38}$ keV. The strong suppression of the latter mode follows from destructive interference between the two physical-current components.

I. INTRODUCTION

The spectroscopy of positive-parity strange heavy mesons remains a useful testing ground for the interplay between heavy-quark symmetry, light-quark dynamics, and nearby hadronic thresholds. The charm-strange states $D_{s1}(2460)$ and $D_{s1}(2536)$ are especially interesting. The lower state is very narrow and has observed radiative decay modes, whereas the upper state is seen clearly in hadronic channels but its radiative transition to $D_s\gamma$ has not been observed with a comparable significance [1–3]. In the bottom-strange sector the state $B_{s1}(5830)$ has been established in hadronic spectroscopy [4], while direct radiative information is still absent.

Radiative widths of these mesons have been considered in quark models, relativistic quark models, effective approaches, and phenomenological analyses [5–15]. A recent observation by Bondar and Milstein is particularly relevant for the present work: the smallness of $D_{s1}(2536) \to D_s\gamma$ can be understood as a cancellation between the two spin components of the physical axial state [14, 15]. One of our aims is to examine the same mechanism in a light-cone QCD sum-rule (LCSR) calculation.

The LCSR method expresses the correlation function near the light cone in terms of photon distribution amplitudes (DAs) and hadronic decay constants [16–18]. It is well suited for exclusive radiative transitions because it keeps the hard photon emission from quark lines and the long-distance photon emission encoded in photon DAs in a single framework. In the present analysis we include both quark-line hard emission terms, the two-particle photon DAs, and the three-particle photon-DA terms induced by the background-gluon insertion in the quark propagator. We also keep the terms proportional to the strange-quark mass.

The paper is organized as follows. In Sec. II we define the currents, the radiative form factor, and the LCSR correlation function. The sum-rule derivation and the physical-current normalization are summarized in Sec. III. Section IV contains the input parameters, Borel-window discussion, numerical results, comparison with other studies, and experimental interpretation. We conclude in Sec. V.

II. THEORETICAL FRAMEWORK

We consider the transition of an axial strange heavy meson with momentum $P = p+q$ into a pseudoscalar strange heavy meson with momentum p and a real photon with momentum q. The pseudoscalar interpolating current is

$$J_5 = \bar{s}\,\mathrm{i}\gamma_5 Q, \qquad Q = c, b. \tag{1}$$

For the axial channel we use the two basis currents

$$J_\mu^A = \bar{s}\gamma_\mu\gamma_5 Q, \tag{2}$$

$$J_\mu^B = \frac{\mathrm{i}}{m_Q + m_s}\,\bar{s}\sigma_{\mu\alpha}P^\alpha\gamma_5 Q, \tag{3}$$

where P^α is the momentum of the initial axial meson. The physical currents are written as

$$J_{1\mu} = \sin\theta\, J_\mu^A + \cos\theta\, J_\mu^B, \tag{4}$$

$$J_{2\mu} = \cos\theta\, J_\mu^A - \sin\theta\, J_\mu^B. \tag{5}$$

Unless stated otherwise we use $\theta = 35.3^\circ$ and include the interval $25^\circ \leq \theta \leq 45^\circ$ as a systematic scan.

The radiative matrix element is parametrized by a single gauge-invariant electric-dipole form factor:

$$\langle P(p)\gamma(q,\varepsilon)|A(P,\eta)\rangle = eG\left[(\eta\cdot\varepsilon^*)(p\cdot q) - (\eta\cdot q)(p\cdot\varepsilon^*)\right]. \tag{6}$$

The corresponding width is

$$\Gamma(A\to P\gamma) = \frac{\alpha_{\mathrm{em}}}{3}\,G^2\left(\frac{m_A^2 - m_P^2}{2m_A}\right)^3. \tag{7}$$

III. LIGHT-CONE SUM RULE

The starting point is the vacuum-to-photon correlation function

$$\Pi_\mu(p,q) = \mathrm{i}\int d^4x\, e^{\mathrm{i}p\cdot x}\langle\gamma(q,\varepsilon)|T\{\bar{s}(x)\Gamma_\mu Q(x)\,\bar{Q}(0)\mathrm{i}\gamma_5 s(0)\}|0\rangle, \tag{8}$$

where $\Gamma_\mu = \gamma_\mu\gamma_5$ or $\Gamma_\mu = \mathrm{i}\sigma_{\mu\alpha}P^\alpha\gamma_5/(m_Q + m_s)$. We project the correlator onto

$$(p \cdot q)\varepsilon^*_\mu - (p \cdot \varepsilon^*)q_\mu, \tag{9}$$

which matches the structure in Eq. (6).

On the hadronic side, isolating the ground-state double pole gives

$$\Pi^{\text{had}}_\mu = \frac{m_P^2 f_P}{m_Q + m_s}\,\frac{m_A f_A^{(i)} G_i}{(m_A^2 - P^2)(m_P^2 - p^2)}\left[(p \cdot q)\varepsilon^*_\mu - (p \cdot \varepsilon^*)q_\mu\right] + \cdots\,, \tag{10}$$

where $i = A, B$ for the basis currents and the ellipsis denotes higher states and continuum contributions.

The QCD side is obtained by expanding near the light cone in the external electromagnetic field. We use the quark propagators in the external photon and gluon background. For the heavy-quark line the relevant terms are

$$\begin{aligned} S_Q(k) = \frac{\not{k} + m_Q}{m_Q^2 - k^2} &- e_Q \int_0^1 du \left[\frac{\not{k} + m_Q}{2(m_Q^2 - k^2)^2}F^{\mu\nu}(ux)\sigma_{\mu\nu} + \frac{ux_\mu F^{\mu\nu}(ux)\gamma_\nu}{m_Q^2 - k^2}\right] \\ &- g_s \int_0^1 du \left[\frac{\not{k} + m_Q}{2(m_Q^2 - k^2)^2}G^{\mu\nu}(ux)\sigma_{\mu\nu} + \frac{ux_\mu G^{\mu\nu}(ux)\gamma_\nu}{m_Q^2 - k^2}\right] + \cdots\,. \end{aligned} \tag{11}$$

The light-quark propagator is used consistently in the same external-field expansion. Ordinary vacuum-condensate terms are not inserted as independent three-point contributions in the photon LCSR. Instead, the long-distance photon emission is described by photon matrix elements such as $\langle\gamma|\bar{s}\sigma_{\mu\nu}s|0\rangle$, whose normalization contains the susceptibility combination $\chi_s\langle\bar{s}s\rangle$ or, equivalently, the modern lattice normalization $f^\perp_{\gamma,s}$. This convention is distinct from the local two-point sum rules for the residues, where the usual SVZ condensates enter the OPE.

For completeness we spell out the invariant amplitudes used in the numerical analysis. The explicit expressions below are written at the symmetric Borel point used in the calculation, $u_0 = M_1^2/(M_1^2 + M_2^2) = 1/2$, with the effective Borel parameter denoted by M^2. We use Rohrwild's notation for the photon DAs: the two-particle functions are denoted by $\phi_\gamma(u)$, $\psi^{(v)}(u)$, $\mathcal{A}(u)$ and $\mathcal{B}(u)$, while the three-particle twist-4 DAs are $\mathcal{S}$, $\widetilde{\mathcal{S}}$ and $\mathcal{T}_{1,\ldots,4}$ [19].

In this notation the axial-current invariant amplitude is decomposed as

$$
\begin{aligned}
\widehat{\Pi}_A = &\int_{(m_c+m_s)^2}^{s_0} ds\, e^{-s/M^2}\rho_A(s) + e_c\, e^{-m_c^2/M^2}\langle\bar{s}s\rangle\left[1 - \frac{m_c m_s}{M^2} + \frac{m_s^2}{2M^2}\left(1+\frac{m_c^2}{M^2}\right)\right] \\
&+ e_s\,\langle\bar{s}s\rangle\left(e^{-m_c^2/M^2} - e^{-s_0/M^2}\right)M^2\chi_s\,\phi_\gamma(u_0) \\
&- e_s\,\langle\bar{s}s\rangle\, e^{-m_c^2/M^2}\left[-\frac{1}{4}\left(\mathcal{A}(u_0)+\mathcal{B}(u_0)\right)\left(1+\frac{m_c^2}{M^2}\right)\right] \\
&+ 2e_s f_{3\gamma} m_c\, e^{-m_c^2/M^2}\Psi^{(v)}(u_0) \\
&+ e_s\,\langle\bar{s}s\rangle\, e^{-m_c^2/M^2}\,\mathcal{I}_{F_1}(u_0).
\end{aligned} \tag{12}
$$

Here

$$
\mathcal{B}(u_0) = -8\int_0^{u_0} du\,(u_0-u)h_\gamma(u), \qquad \Psi^{(v)}(u_0) = \int_0^{u_0} du\,\psi^{(v)}(u). \tag{13}
$$

The first relation is only a notation dictionary: it rewrites the two-particle twist-4 combination used in the numerical code, $\mathcal{A} - 8\bar{H}_\gamma$, as Rohrwild's $\mathcal{A}+\mathcal{B}$. and the three-particle photon-DA integral is

$$
\begin{aligned}
\mathcal{I}_{F_1}(u_0) = &\int_0^{1-u_0} dv \int_0^{u_0/(1-v)} d\alpha_g\, F_1(u_0-(1-v)\alpha_g, 1-u_0-v\alpha_g, \alpha_g, v) \\
&+ \int_{1-u_0}^{1} dv \int_0^{(1-u_0)/v} d\alpha_g\, F_1(u_0-(1-v)\alpha_g, 1-u_0-v\alpha_g, \alpha_g, v),
\end{aligned} \tag{14}
$$

with

$$
F_1 = \mathcal{S} + \widetilde{\mathcal{S}} - \mathcal{T}_1 - \mathcal{T}_2 + \mathcal{T}_3 + \mathcal{T}_4 + 2v(-\mathcal{S} - \mathcal{T}_3 + \mathcal{T}_2). \tag{15}
$$

The perturbative spectral density in Eq. (12) is

$$
\rho_A(s) = e_s\,\mathcal{R}(s; m_s, m_c; m_s, m_s^2(m_c-m_s)) - e_c\,\mathcal{R}(s; m_c, m_s; m_c, m_c^2(m_s-m_c)), \tag{16}
$$

where

$$
\begin{aligned}
\mathcal{R}(s; m_i, m_j; a, b) = -\frac{3}{8\pi^2}\Bigg[&2a\ln\frac{s-m_j^2+m_i^2-\lambda^{1/2}(s,m_j^2,m_i^2)}{s-m_j^2+m_i^2+\lambda^{1/2}(s,m_j^2,m_i^2)} \\
&+\frac{b}{m_i^2}\,\frac{m_j^2-m_i^2-s}{s^2}\,\lambda^{1/2}(s,m_j^2,m_i^2)\Bigg],
\end{aligned} \tag{17}
$$

and $\lambda(s, m_j^2, m_i^2) = [s-(m_i+m_j)^2][s-(m_i-m_j)^2]$.

The tensor-current invariant amplitude is written in the same notation as

$$
\widehat{\Pi}_B = \int_{(m_c+m_s)^2}^{s_0} ds\, e^{-s/M^2}\rho_B(s) + \frac{m_c}{m_c+m_s}\,\widehat{\Pi}_A^{(\chi\phi_\gamma,\mathcal{A},\mathcal{B})} + \widehat{\Pi}_B^{(3p)}. \tag{18}
$$

The second term denotes the tensor two-particle photon-DA family inherited from the axial expression in Eq. (12), namely the $\chi_s\phi_\gamma$ and $\mathcal{A}, \mathcal{B}$ pieces. The perturbative tensor spectral density is

$$\begin{aligned}\rho_B(s) &= e_s\,\mathcal{R}\left(s; m_s, m_c; \frac{m_c m_s}{m_c+m_s}, \frac{m_s^2 s}{m_c+m_s}\right)\\ &\quad - e_c\,\mathcal{R}\left(s; m_c, m_s; \frac{2m_c^2 - m_c m_s}{m_c+m_s}, -\frac{m_c^2 s}{m_c+m_s}\right).\end{aligned} \tag{19}$$

The last term in Eq. (18), $\widehat{\Pi}_B^{(3p)}$, is the matched tensor-current three-particle contribution generated by the background-gluon part of the propagator. In the implementation it is expressed through the same integration domains as Eq. (14), with the tensor kernels obtained from the $\sigma_{\alpha\beta}$ and $x_\alpha\gamma_\beta$ parts of Eq. (11). The compact representation used numerically is

$$\widehat{\Pi}_B^{(3p)} = e_s\,\langle\bar{s}s\rangle\, e^{-m_c^2/M^2}\,\mathcal{I}_{3p}^B(u_0), \qquad \mathcal{I}_{3p}^B(1/2) = 5.98\times10^{-2}, \tag{20}$$

for the central photon-DA parameters. The corresponding axial integral is $\mathcal{I}_{F_1}(1/2) = 7.03\times10^{-2}$.

After the double Borel transformation in p^2 and P^2 and continuum subtraction, we denote by $\widehat{\Pi}_A(M_1^2, M_2^2, s_0)$ and $\widehat{\Pi}_B(M_1^2, M_2^2, s_0)$ the invariant QCD amplitudes multiplying the structure in Eq. (9) for the two basis currents. The corresponding basis form factors satisfy

$$\begin{aligned}G_A &= \frac{m_Q+m_s}{m_P^2 f_P\, m_A f_A}\exp\left(\frac{m_A^2}{M_1^2}+\frac{m_P^2}{M_2^2}\right)\widehat{\Pi}_A(M_1^2, M_2^2, s_0),\\ G_B &= \frac{m_Q+m_s}{m_P^2 f_P\, m_A f_B}\exp\left(\frac{m_A^2}{M_1^2}+\frac{m_P^2}{M_2^2}\right)\widehat{\Pi}_B(M_1^2, M_2^2, s_0).\end{aligned} \tag{21}$$

Here m_A is the mass of the axial state used in the corresponding hadronic pole, m_P and f_P are the mass and decay constant of the final pseudoscalar meson, and M_1^2 and M_2^2 are the Borel parameters in the axial and pseudoscalar channels, respectively.

For the physical charm-strange states, the sum rules for the form factors are therefore

$$\begin{aligned}G_1 \equiv G_{2460} &= \frac{m_c+m_s}{m_{D_s}^2 f_{D_s}\, m_1 f_1}\exp\left(\frac{m_1^2}{M_1^2}+\frac{m_{D_s}^2}{M_2^2}\right)\left[\sin\theta\,\widehat{\Pi}_A^{(2460)} + \cos\theta\,\widehat{\Pi}_B^{(2460)}\right],\\ G_2 \equiv G_{2536} &= \frac{m_c+m_s}{m_{D_s}^2 f_{D_s}\, m_2 f_2}\exp\left(\frac{m_2^2}{M_1^2}+\frac{m_{D_s}^2}{M_2^2}\right)\left[\cos\theta\,\widehat{\Pi}_A^{(2536)} - \sin\theta\,\widehat{\Pi}_B^{(2536)}\right],\end{aligned} \tag{22}$$

where $m_1 = m_{D_{s1}(2460)}$ and $m_2 = m_{D_{s1}(2536)}$. Equivalently, if the basis-current form factors in Eq. (21) are introduced first, the same physical couplings are written as

$$G_{2460} = \frac{\sin\theta\, f_A\, G_A + \cos\theta\, f_B\, G_B}{f_1}, \tag{23}$$

$$G_{2536} = \frac{\cos\theta\, f_A\, G_A - \sin\theta\, f_B\, G_B}{f_2}. \tag{24}$$

TABLE I. Main input parameters. Photon-DA normalizations are quoted at $\mu = 1$ GeV.

Quantity	Value	Comment
m_c, m_b, m_s	1.27(2), 4.18(3), 0.093(11) GeV	$\overline{\text{MS}}$ inputs
$M_{2460}, M_{2536}, m_{D_s}$	2.4595, 2.53511, 1.96835 GeV	PDG
M_{5830}, m_{B_s}	5.82870, 5.36692 GeV	PDG/CMS
f_{D_s}, f_{B_s}	0.2499, 0.2303 GeV	FLAG/lattice context
f_1, f_2	0.345(17), $\simeq 0.38$ GeV	charm mixed-current residues
$f_{B_{s1}}, f^T_{B_{s1}}$	0.305(27), 0.285(27) GeV	bottom PZ scenario
$\chi_s(1\text{ GeV})$	$+3.15(30)$ GeV^{-2}	legacy photon input
$f^{\perp}_{\gamma,s}(1\text{ GeV})$	$-55.1(1.9)$ MeV	preferred photon input

Here f_1 and f_2 are the residues of the mixed physical currents in Eq. (5). We use a Stage-3 residue normalization in which $f_1 = 0.345 \pm 0.017$ GeV is used as the working anchor and the same two-point overlap model gives $f_2 \simeq 0.38$ GeV. As a check of the basis normalization, the pure axial-current limit is compared with the QCD sum-rule result of Ref. [20]. For the bottom-strange channel we use the Pullin–Zwicky decay-constant scenario $f_{B_{s1}} = 0.305 \pm 0.027$ GeV and $f^T_{B_{s1}} = 0.285 \pm 0.027$ GeV [21].

IV. NUMERICAL ANALYSIS

A. Input parameters and working windows

The main numerical inputs are collected in Table I. Masses and quark masses are taken from the Particle Data Group [22]. The pseudoscalar decay constants are taken from the lattice averages summarized by FLAG [23]. The photon DAs are those of Ball, Braun and Kivel [18]. We compare the older magnetic-susceptibility input [19] with the recent lattice normalization of the leading-twist photon DA [24]; the lattice-normalized scenario is used as our preferred result.

The Borel windows are chosen by requiring a stable prediction under changes of M^2 and the continuum threshold while avoiding regions where continuum contributions or higher-dimensional terms dominate. For the charm-strange case we use $3\text{ GeV}^2 \leq M^2 \leq 6\text{ GeV}^2$ and

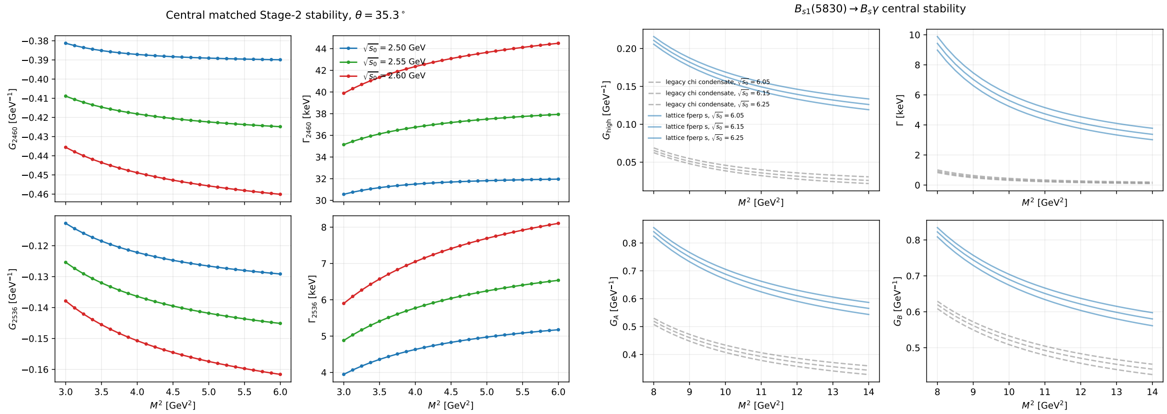

FIG. 1. Representative Borel and continuum-threshold stability checks for the charm-strange and bottom-strange channels.

TABLE II. Final form-factor and width predictions. The lattice photon input is our preferred normalization.

Sector	Decay	Photon input	Normalization	G (GeV^{-1})	Γ (keV)
D_s	$D_{s1}(2460) \to D_s\gamma$	legacy χ_s	$f_1 = 0.344[0.329, 0.363]$	$-0.424[-0.484, -0.360]$	$37.7^{+11.5}_{-10.5}$
D_s	$D_{s1}(2536) \to D_s\gamma$	legacy χ_s	$f_2 = 0.380[0.339, 0.423]$	$+0.0118[+0.0014, +0.0233]$	$0.045^{+0.124}_{-0.040}$
D_s	$D_{s1}(2460) \to D_s\gamma$	lattice $f^{\perp}_{\gamma,s}$	$f_1 = 0.345[0.330, 0.364]$	$-0.277[-0.312, -0.243]$	$16.1^{+4.4}_{-3.7}$
D_s	$D_{s1}(2536) \to D_s\gamma$	lattice $f^{\perp}_{\gamma,s}$	$f_2 = 0.379[0.338, 0.423]$	$+0.0403[+0.0194, +0.0596]$	$0.504^{+0.600}_{-0.377}$
B_s	$B_{s1}(5830) \to B_s\gamma$	legacy χ_s	PZ constants	$-0.0140[-0.0387, +0.0155]$	$0.109^{+0.304}_{-0.101}$
B_s	$B_{s1}(5830) \to B_s\gamma$	lattice $f^{\perp}_{\gamma,s}$	PZ constants	$+0.0356[+0.0029, +0.0694]$	$0.272^{+0.752}_{-0.244}$

2.50 GeV $\leq \sqrt{s_0} \leq 2.60$ GeV. For the bottom-strange case we use $8\ \text{GeV}^2 \leq M^2 \leq 14\ \text{GeV}^2$. The representative stability plots are shown in Fig. 1.

B. Results and uncertainties

The final form factors and widths are listed in Table II. We quote medians and 16–84 percentile intervals. This is preferable to a Gaussian fit for cancellation-sensitive channels, because the distributions of $D_{s1}(2536)$ and $B_{s1}(5830)$ are visibly non-Gaussian near zero.

The dominant uncertainty in the legacy scenario comes from the photon susceptibility normalization and the condensate combination entering $\chi_s\langle\bar{s}s\rangle$. Replacing this input by the lattice-normalized $f^{\perp}_{\gamma,s}$ reduces the photon-normalization uncertainty and is the reason why we regard the lattice rows as the preferred modern prediction. The mixing angle is the

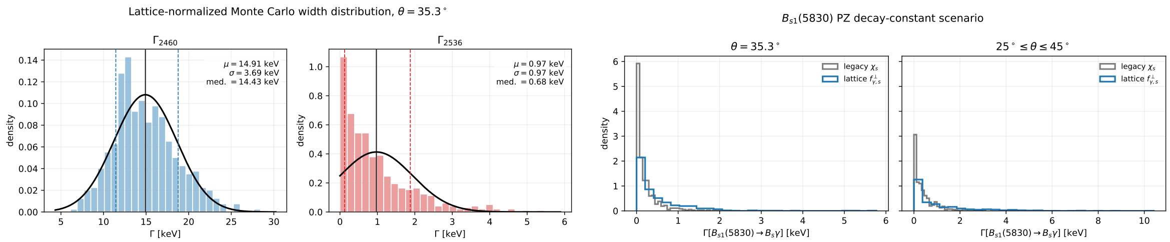

FIG. 2. Monte Carlo width distributions. Left: charm-strange lattice-normalized photon input. Right: bottom-strange PZ decay-constant scenario.

TABLE III. Comparison with representative theoretical predictions. Widths are in keV.

Channel	Quark models	Recent QM	Bondar–Milstein	This work
$D_{s1}(2536) \to D_s\gamma$	15, 25.2–31.1	18.18–18.85	12	$0.504[0.127, 1.104]$
$D_{s1}(2460) \to D_s\gamma$	6.2, 10.3–17.2	3.53–3.61	297	$16.1[12.4, 20.5]$
$B_{s1}^{\mathrm{low}}(5750) \to B_s\gamma$	37, 70.6	97.7	47	$27.8[20.1, 39.4]$
$B_{s1}(5830) \to B_s\gamma$	27, 47.8	56.6	28	$0.272[0.028, 1.024]$

other important source of uncertainty, especially for the cancellation-sensitive $D_{s1}(2536)$ and $B_{s1}(5830)$ amplitudes. The Monte Carlo distributions for the preferred charm and bottom scenarios are shown in Fig. 2.

The suppression pattern can be seen directly at the amplitude level. With the lattice photon normalization, the normalized pieces of the $D_{s1}(2460)$ amplitude are -0.0715 and -0.205, and therefore add constructively. For $D_{s1}(2536)$ the corresponding pieces are -0.0917 and $+0.132$, so the physical amplitude is a small difference. This is the LCSR counterpart of the cancellation mechanism emphasized in Refs. [14, 15].

C. Comparison with other studies and experiment

Table III compares our preferred values with representative predictions in the literature. The comparison includes early quark-model estimates, relativistic quark-model calculations, recent potential-model or electric-transition analyses, and the phenomenological results of Bondar and Milstein. We compute the pseudoscalar final-state channels $D_s\gamma$ and $B_s\gamma$. The vector-final-state channels $D_s^*\gamma$ and $B_s^*\gamma$ require separate LCSR correlators and are not part of this paper.

For $D_{s1}(2460)$, using the measured branching fraction $\mathcal{B}[D_{s1}(2460) \to D_s\gamma] = (16 \pm 4 \pm 3)\%$ [1] with our preferred partial width implies a total-width scale of order 0.10 MeV, compatible with a very narrow state. For $D_{s1}(2536)$, the measured total width $\Gamma_{\rm tot} = 0.92 \pm 0.05$ MeV [2] implies a radiative branching fraction of order 5×10^{-4} in our preferred scenario. This is far below the percent level and naturally explains why the decay is difficult to isolate experimentally. For $B_{s1}(5830)$ there is not yet a direct radiative measurement, and our result should be read as a prediction for future searches.

V. CONCLUSION

We have calculated the radiative transitions $D_{s1}(2460) \to D_s\gamma$, $D_{s1}(2536) \to D_s\gamma$, and $B_{s1}(5830) \to B_s\gamma$ using light-cone QCD sum rules. The calculation includes hard photon emission, two- and three-particle photon DAs, and explicit strange-quark-mass terms. We separated the basis-current sum rules from the physical-current residue normalization and used a lattice-normalized photon DA as the preferred modern input.

The main result is the strong hierarchy between the two charm-strange radiative widths. The $D_{s1}(2460)$ transition is naturally at the tens-of-keV level, while $D_{s1}(2536) \to D_s\gamma$ is strongly suppressed by destructive interference between the two physical-current components. The same cancellation-sensitive pattern appears for the observed $B_{s1}(5830)$, for which we predict a sub-keV median radiative width.

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To be added.

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